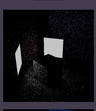
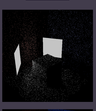
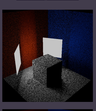
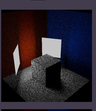
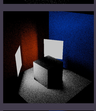
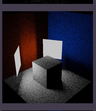
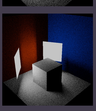
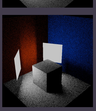
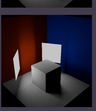
Direct Lighting
*****Lambert***

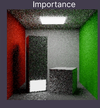
	Uniform	PDF Cosine	Area importance	MIS*
1 Ray				
50 Rays				
128 Rays				
256 Rays				

* MIS is combining Area Importance + PDF Cosine and Area Importance is the workhorse while the PDF Cosine contribution is marginal, hence the low gain compared with Area Importance sampling
** MIS Area Importance + PDF Cosine works well with indirect lighting as seen bellow in Lambert + GGX, 1 bounce Cornell Box

10 Cosine light bounces

10 Area Importance no bounce

10 MIS from Cosine and Importance

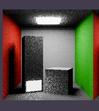


Area uniform
left small light is over powered

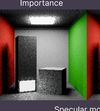
Area importance
w = Σ*Ai*Li
Left light is all right

Resampled importance sampling
10 rays direct Lambert + GGX

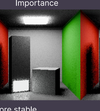
10 Area importance



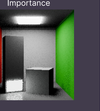
10 RIS x 2 from Importance



10 RIS x 8 from Importance



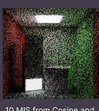
10 RIS x 32 from Importance



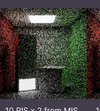
Specular more stable

10 rays 1 bounce, Lambert + GGX

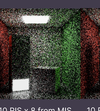
10 Cosine



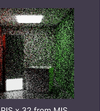
10 RIS x 2 from Cos



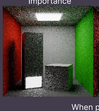
10 RIS x 8 from Cos



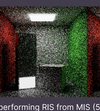
10 RIS x 32 from Cos



10 MIS from Cosine and Importance

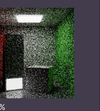


10 RIS x 2 from MIS

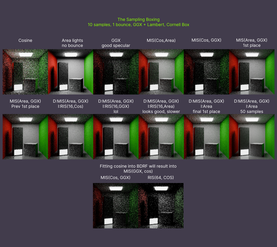


10 RIS x 8 from MIS

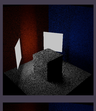
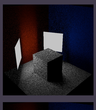
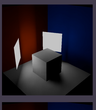
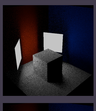
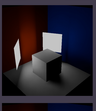
10 RIS x 32 from MIS



When performing RIS from MIS (50% Area Importance, 50 % Cosine), mostly Cosine survives the collapsing



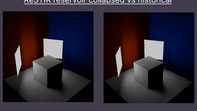
Resampled sampling
ReSTIR-lite, one ray per pixel, direct lighting, Lambert
(lite means no RIS, historical data is used)

	0×0 Spatial Kernel	7×7 Spatial Kernel	15×15 Spatial Kernel
Temporal budget 1 No history			
Temporal budget 2			
Temporal budget 4			
Temporal budget 8			

Spatial samples drop

	0% samples dropped	25% samples dropped	50% samples dropped	90% samples dropped
Temporal budget 1 7 × 7 Spatial Kernel				
Temporal budget 4 7 × 7 Spatial Kernel				
Temporal budget 8 7 × 7 Spatial Kernel				

ReSTIR reservoir collapsed vs historical



ReSTIR spatial vs temporal

ReSTIR Historical

Maximum of 8 temporal samples are kept in the reservoir, allowing recomputation of PDF and cos θ.

ReSTIR pure

Collapsed reservoir. Only the reservoir champion sample is kept.

Spatial merging of 50 samples with visibility resampled

Temporal budget 1 7×7 Spatial Kernel

Temporal merging of 50 samples

Temporal budget 50 0×0 Spatial Kernel

No major differences noticed

Temporal resampling dominates